Comparison of
Simple Lossless Encoding Vs. Run-Length Encoding (RLE)

Aspect	Simple Lossless Encoding	Run-Length Encoding (RLE)
Definition	Any compression method that allows the exact original data to be reconstructed (no loss of information). Examples: Huffman coding, LZW, arithmetic coding.	A specific type of lossless encoding that compresses data by storing runs of repeated symbols as a single symbol and its count.
Encoding Principle	Uses statistical models or dictionaries to represent frequent symbols/phrases with shorter codes.	Counts consecutive identical symbols and stores (symbol, run-length) pairs.
Mathematical Model	Compression ratio depends on symbol probability distribution P(x). For an optimal code: $L\_avg \geq H(X)$, where $H(X) = - \Sigma P(x) \log_2 P(x)$ (Shannon entropy).	Compression ratio depends on run-length distribution: $CR = (n \times b) / (r \times (b + \log_2 R))$, where n = original symbols, b = bits per symbol, r = number of runs, R = max run length.
When Efficient	Works best when certain symbols occur more frequently (high statistical redundancy).	Works best when data has long runs of identical symbols (high spatial redundancy).
Example Encoding	Text: "ABABABAB" → Huffman code may map A=0, B=1 → 01010101.	Text: "AAAABBBCC" → (A,4)(B,3)(C,2).
Space Efficiency	Compression gain comes from variable-length codes for common symbols.	Compression gain comes from reducing repeated symbols to (symbol, count) representation.
Mathematical Limitation	Cannot exploit redundancy if symbol probabilities are uniform.	Inefficient if run lengths are short — may even increase size.
Algorithmic time complexity	Encoding/decoding complexity depends on coding algorithm, often $O(n\log_2 n)$ for Huffman.	Encoding/decoding complexity is O(n) since it is a linear pass-through data.
Application Examples	ZIP files, PNG images, text compression.	Bitmap images with uniform color areas, simple icons, monochrome faxes.
Lossless Property (Common)	Always lossless — original data reconstructed exactly.	Always lossless — original data reconstructed exactly.